

ZIDIO DEVELOPMENT · DATA SCIENCE & ANALYTICS

Project FORESIGHT

AI-powered demand & inventory intelligence for NorthBay Living

Project report

Client: NorthBay Living (D2C home & lifestyle)

Role: Data Scientist

Stack: Python · pandas · LightGBM · Streamlit · FastAPI

Data: Jan 2024 - Dec 2025, 50 SKUs, stock as of 01 Dec 2025

1. Overview

NorthBay Living plans inventory on gut feel and spreadsheets. It loses money both ways: best-sellers run out and slow movers pile up. The client asked for four things: a demand forecast, a stockout early warning, an overstock flag, and something a non-technical team can use. FORESIGHT delivers all four.

- A weekly, SKU-level demand forecast for the next **8 weeks** with an 80% likely range. It beats a seasonal-naive baseline on a rolling-origin backtest.
- Stockout and overstock risk for every SKU, with a recommended action and the rupee value at stake.
- A Streamlit planning dashboard and a FastAPI scoring service, both fed by one reproducible pipeline.
- A data-quality & EDA memo and an executive readout for the Head of Operations and Finance.

Headline result	Value
Backtest WAPE: FORESIGHT model	8.7%
Backtest WAPE: seasonal-naive baseline	10.5%
Improvement over baseline · folds won	17.6% · 10/10
Hold-out after stock date (4 weeks): model vs baseline	8.7% vs 12.4%
Bias (model)	-0.9%
SKUs to reorder now · sales at risk	7 · ₹2.78 Cr
SKUs to mark down / clear · cash locked	4 · ₹32.9 L

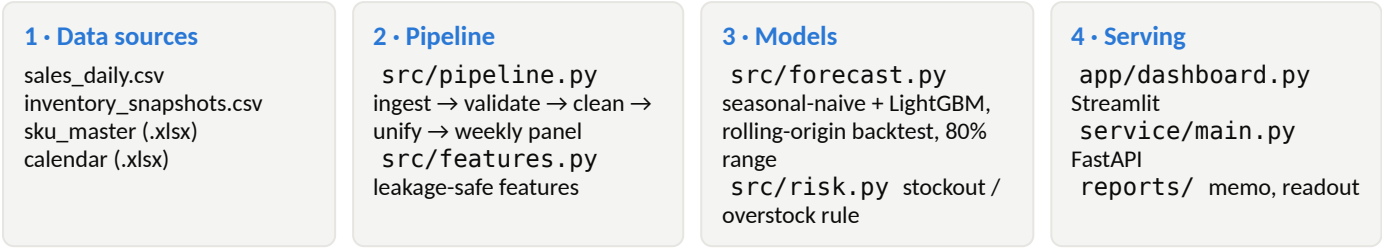
2. Data

Four extracts from the client: **sales_daily** (36,550 rows, one per SKU per day), **sku_master** (50 SKUs), **calendar** (731 days, holidays and promotion events) and **inventory_snapshots** (monthly, 200 SKUs). The main data issues, each handled in code and logged: an unlabelled cost column (`wfr` , verified as unit cost), 150 stock-only SKUs, launch dates after first sales, mismatched subcategories, stock values that don't equal stock × cost, snapshots that don't reconcile with sales, and 16 SKUs priced below cost. Full detail is in the EDA memo and `outputs/data_quality_log.csv` .

3. Tech stack

Area	Tools
Language & analysis	Python 3.11, pandas, NumPy, Jupyter
Modelling	LightGBM (via its scikit-learn interface)
Visualisation	matplotlib (reports), Plotly (dashboard)
Dashboard	Streamlit
Scoring service	FastAPI + Uvicorn, Pydantic validation
Reports	HTML rendered to PDF with Chromium (Playwright)
Testing	pytest (leakage, risk rules, API error handling)
Deployment	Streamlit Community Cloud (dashboard), Render (API, render.yaml blueprint)
Version control	Git

4. Architecture



Repository layout

data/raw · data/processed	client extracts as delivered · cleaned, analysis-ready tables
src/	config, pipeline, features, forecast, risk, eda
notebooks/	01_ed, 02_baseline, 03_model (executed)
app/ · service/	Streamlit dashboard · FastAPI scoring service
outputs/	forecast, risk scores, backtest results, metrics, data-quality log
reports/ · tests/	PDF deliverables and figures · pytest checks

5. Methodology

Frame and baseline

The target is weekly units per SKU (Monday-Sunday) over an 8-week horizon. The primary metric is WAPE, with bias as a secondary check. The baseline is seasonal-naive: each week is forecast as the same week last year.

Features and model

One global LightGBM model covers all SKUs and predicts each horizon directly (one row per SKU × origin × horizon). Features: recent lags, 4/8/13/26-week rolling means, volatility, same-week-last-year (raw, smoothed and adjusted for level), planned promotion and holiday days in the target week, week of year, month, category and price. Demand features are scaled by each SKU's recent level. The model is trained with an absolute-error loss weighted by that level, so it minimises the same quantity WAPE measures.

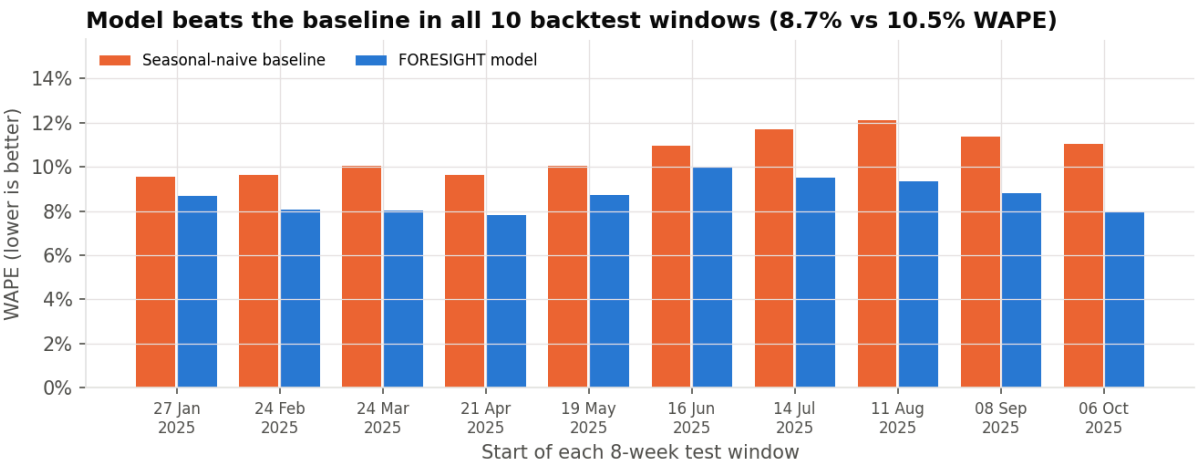
Backtest and leakage control

The backtest uses rolling origins: 10 origins, four weeks apart, from Jan to Sep 2025. Each fold trains only on weeks up to its origin and forecasts the next 8. Features use only data up to the origin; only calendar facts known in advance describe the target week. A unit test scrambles all future demand and confirms that no feature changes. As an extra check, the 4 weeks after the stock date were never used for training or model choice.

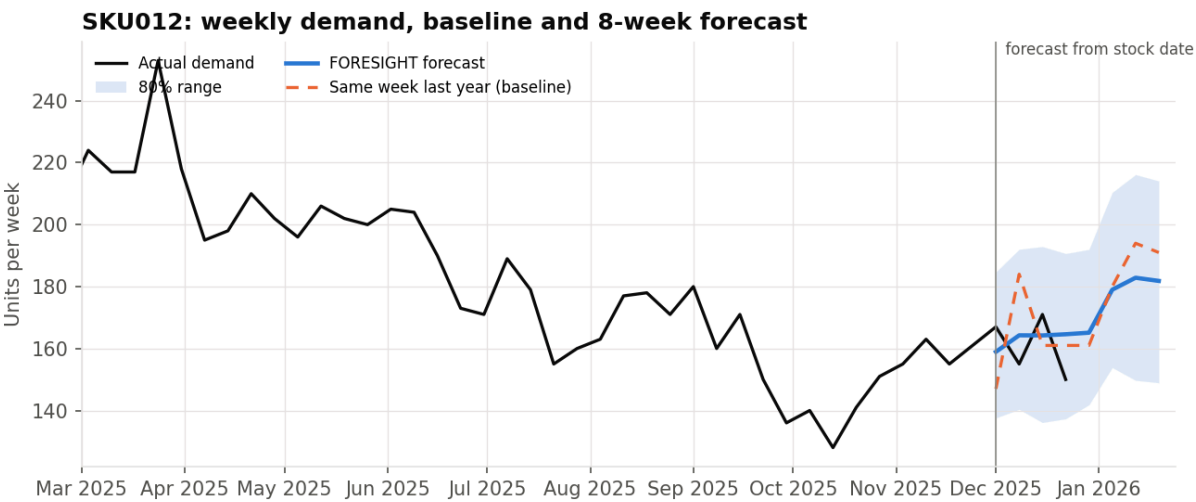
Risk scoring

Stockout risk is the probability that lead-time demand exceeds on-hand + on-order stock minus safety stock. It is flagged above 10%, which corresponds to a 90% service level. **Overstock risk** is the probability that 8-week demand is below on-hand stock minus safety stock, flagged above 50%. Demand uncertainty comes from the empirical backtest error for each window length, scaled by the SKU's own forecast reliability. The two flags map to the four quadrants of the Section 08 grid. Each SKU also gets a value at stake: sales at risk (revenue the stock cannot serve) or cash locked (excess stock at cost).

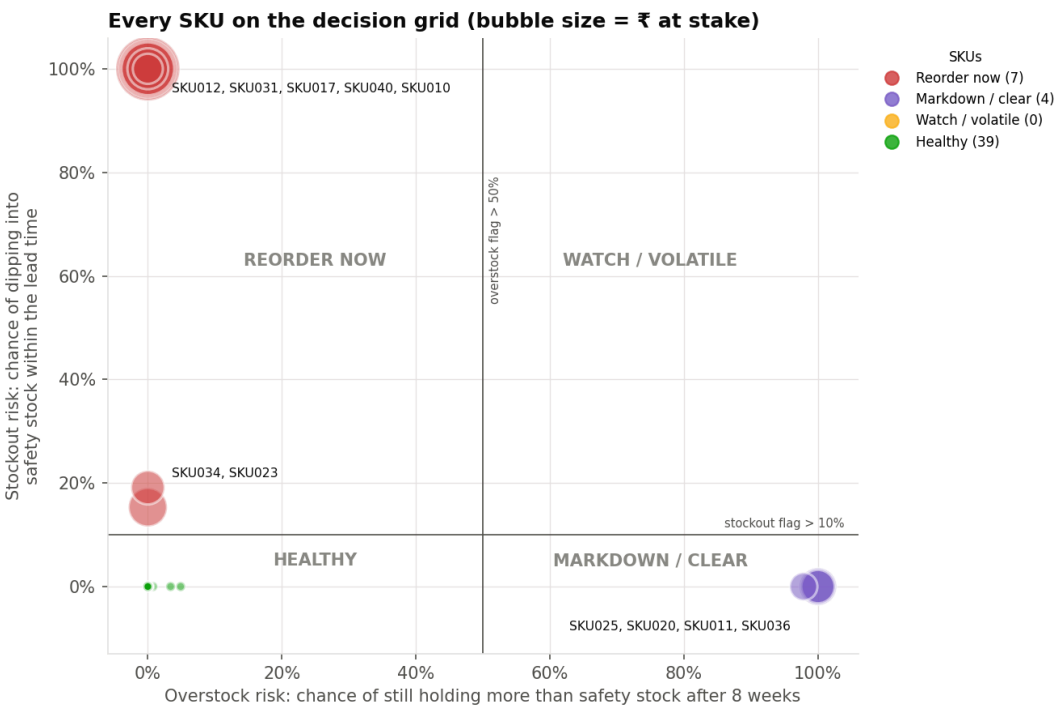
6. Results



WAPE for each backtest window: the model beats the baseline in every one.



Forecast output for the highest-value SKU: actual history, baseline, forecast and 80% range.



Decision grid: 7 reorder now, 4 markdown / clear, 0 watch / volatile, 39 healthy.

7. Dashboard and scoring service

The dashboard gives the operations team four views: **action lists** (prioritised reorder and markdown tables with rupee values), **forecast vs actual** per SKU (history, past backtest forecasts, the 8-week forecast with its range, and a live check against weeks that have already happened), the **risk grid**, and a plain-language **how to read this** page. Filters for category, status and SKU sit in the sidebar. Empty filters show a clear message, and a missing pipeline run shows setup instructions.

Filters

Category

Furniture x

Home Decor x

Kitchen x

Lighting x

Storage x

Status

Reorder now x

Markdown / clear x

Watch / volatile x

Healthy x

SKU

All SKUs

Stock position as of 01 Dec 2025 (latest snapshot). Forecast covers 2025-12-01 to 2026-01-19 (8 weeks).

Deploy

FORESIGHT - what to reorder, what to clear

NorthBay Living · 50 SKUs · stock as of 01 Dec 2025 · next 8 weeks

SKUs to reorder now

7

Sales at risk if not reordered

₹2.78 Cr

SKUs to mark down / clear

4

Cash locked in excess stock

₹32.9 L

Action lists

Forecast vs actual

Risk grid

How to read this

Reorder now - highest sales at risk first

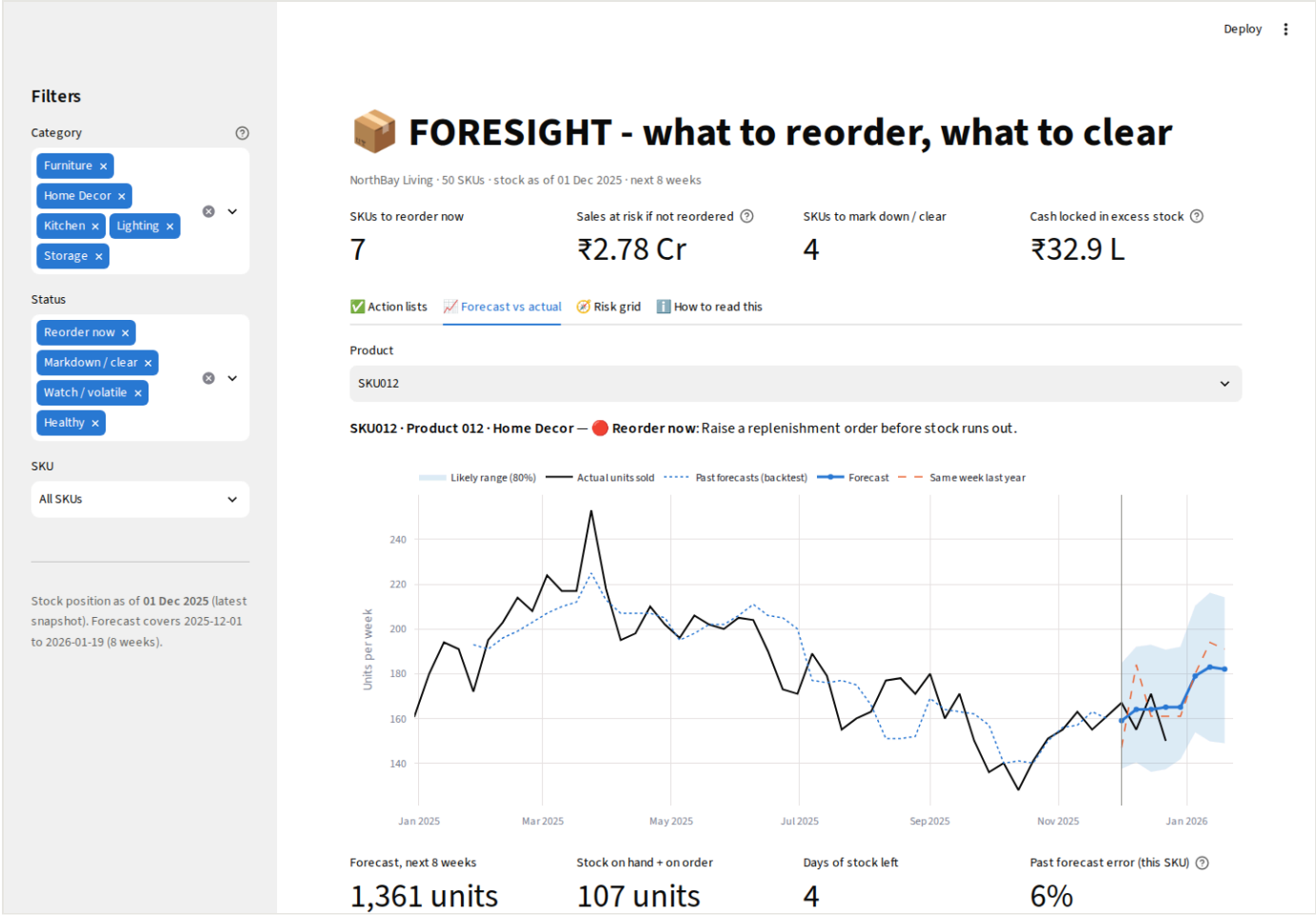
SKU	Product	Sales at risk (₹)	Suggested order (units)	Stockout risk	Days of stock left	Lead time (days)	On hand	On order	Forecast
SKU012	Product 012	11,007,845	1270	100%	4.4	12	47	60	
SKU031	Product 031	7,372,921	874	100%	7.6	13	93	40	
SKU017	Product 017	5,097,239	710	100%	9.4	12	95	26	
SKU040	Product 040	1,749,172	750	100%	6.6	9	67	29	
SKU034	Product 034	1,369,801	937	15.32%	7.4	5	34	103	
SKU023	Product 023	765,445	582	19.12%	15.6	12	130	79	
SKU010	Product 010	468,645	729	99.99%	10	13	83	70	

Suggested order = units needed to cover forecast demand for the next 8 weeks and restore safety stock. It is a recommendation for the planner - FORESIGHT does not place orders.

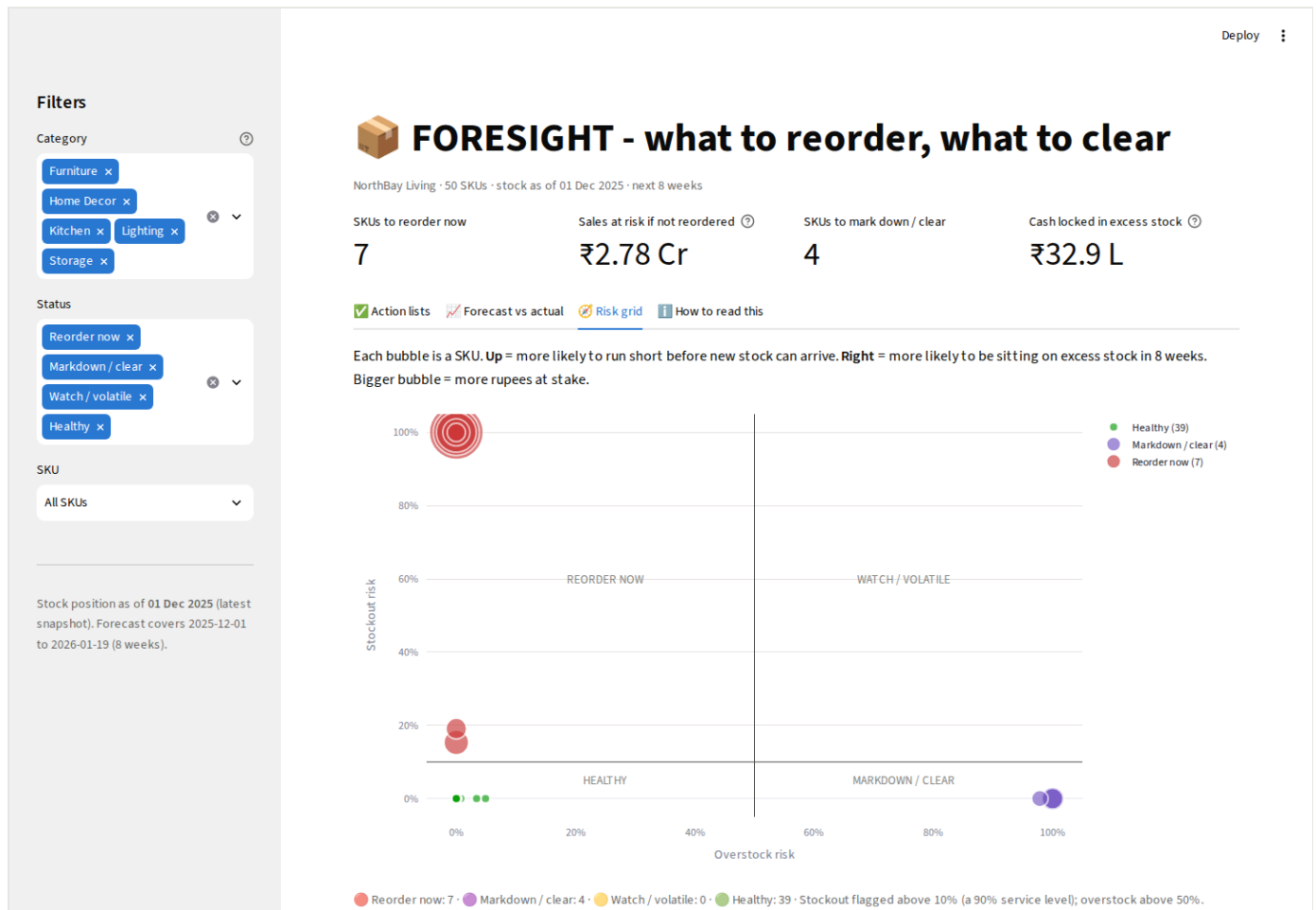
Mark down / clear - most cash locked first

SKU	Product	Cash locked (₹, at cost)	Excess units	Overstock risk	On hand	Forecast next 8 wks	Days of stock left	Category
SKU025	Product 025	1,149,909	863	100%	1124	140	472.6	Storage
SKU020	Product 020	1,080,286	161	99.99%	1002	712	106.6	Storage
SKU011	Product 011	744,185	433	100%	753	141	406.1	Furniture

Action lists with headline KPIs.



Forecast vs actual for one SKU.



Risk grid (interactive, with a tooltip per SKU).

Scoring service (FastAPI)

Endpoint	Returns
GET /health	status and number of SKUs loaded
GET /skus	valid SKU ids
GET /score/{sku_id}	8-week forecast with range, plus risk, action and rupees at stake
POST /score	the same for a batch of up to 200; optional fresher stock inputs per SKU are re-scored with the same rule

Bad input is handled without crashing: an unknown SKU returns 404 (or a per-item error inside a batch), and a negative or non-numeric value returns 422 with a readable message. Interactive documentation is served at `/docs`.

8. Reproducibility and deployment

- `pip install -r requirements.txt` then `python run_pipeline.py` rebuilds every output from the raw files in about a minute, with fixed seeds and single-threaded training so the numbers are identical on every run.
- Dashboard: Streamlit Community Cloud, entry point `app/dashboard.py`. Scoring service: Render, using `render.yaml`. Both read the seeded outputs committed in `outputs/`.

9. Limitations

- Stock snapshots are monthly and don't reconcile with sales, so risk is scored as of the latest snapshot.
- Recorded unit costs exceed price for 16 SKUs, which affects cash-locked values until Finance confirms them.
- 150 stock-only SKUs cannot be forecast.

- Promotion days for January 2026 are assumed to follow the historical pattern.
- Suggested order sizes ignore supplier minimums and pack sizes.

10. Conclusion

FORESIGHT turns NorthBay's own extracts into a forecast that is measurably better than the "same as last year" rule (8.7% vs 10.5% average miss, better in every test window). It also produces a transparent monthly action list. This month that means reordering 7 SKUs to protect ₹2.78 Cr of sales and clearing 4 to free ₹32.9 L of cash. The pipeline re-runs from raw data with one command, so the team can refresh it every month without a data scientist in the room.